

DenseNet-Based Robust Channel Estimation in OFDM for Improving Underwater Acoustic Communication

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I. INTRODUCTION

Abstract—Underwater acoustic (UWA) communication presents unique challenges due to the unpredictable and dynamic nature of acoustic channels, influenced by Doppler spread, low signal-to-noise ratios (SNRs), and the general need for complex channel characteristics, coupled with a scarcity of real-world data. Accurate orthogonal frequency division multiplexing (OFDM) channel estimation is pivotal for ensuring reliable data transmission in such challenging environments. In this study, we introduce the DenseNet estimator, which is specifically used for OFDM channel estimation in UWA communication. The use of dense connectivity within the DenseNet structure proves to be advantageous in capturing the intricacies of the complex and dynamic UWA channels. This architecture, showcasing robustness even when there's a limited number of pilots, sets it apart from conventional methods. The DenseNet estimator is trained on the WATERMARK data set, leveraging the richness of real-time varying channel impulse responses to provide the necessary diversity for accurate channel estimation. Uniquely, once trained, our DenseNet estimator operates without necessitating additional channel statistics like SNR, relying solely on the received signal as its primary input. This approach offers a simplified and more direct application in real-world scenarios. Our numerical results underscore the DenseNet estimator's efficacy: It consistently outperforms traditional methods such as least square, minimum mean square error, and fully connected neural network, recording improvements of up to 96.3%, 94.2%, and 40.0% in bit error rate. Performance assessments across various watermark underwater channels demonstrate the DenseNet estimator's adaptability and robustness in both stable and challenging environments.

Index Terms—Channel estimation, modulation, neural networks, orthogonal frequency division multiplexing (OFDM), underwater acoustic (UWA) communication.

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UNDERWATER wireless communication systems are increasingly popular due to their importance in underwater environment monitoring, undersea rescues, and deep-sea mining. However, these systems must overcome significant obstacles to ensure reliable and fast communication. The propagation of acoustic waves is slow, there is significant propagation loss, and reflections affect acoustic signals. They provide several signal paths with significant delays as they reflect off the surface and bottom of the water [1]. The delay in propagation renders them susceptible to the Doppler effect making the underwater channel time-varying doubly selective for underwater acoustic (UWA) communication [2]. Compared to terrestrial radio frequency-based systems, UWA communication is difficult to perform reliably due to these limits.

When UWA communication systems operate under very low signal-to-noise ratios (SNRs), the signals become weak. In this regard, multicarrier transmission techniques such as orthogonal frequency division multiplexing (OFDM) have attracted interest to improve communication stability in these challenging underwater channels [3]. One such technique that is being used especially for to improve the performance of underwater communication system [4] are OFDM with cyclic prefix (CP) [5]. However, CP-OFDM has inherent advantages compared to the single-carrier channels. It is particularly effective in environments with multipath propagation and offers resistance to degradation of the signal. In addition, it is spectrally efficient and cost-effective implementations of transceivers.

Channel estimation in UWA is crucial to achieve reliable transmission [4], since accurate knowledge of the channel at the receiver side is a necessary condition for successful data reception and widely applied communication systems. Since the CSI is essential for signal detection, pilot-assisted channel estimation technique is mainly used to obtain it. In this approach, a set of prearranged pilot symbols is typically sent alongside the data subcarriers. These pilot symbols serve as reference points to assist in the estimation of the channel's characteristics, ultimately contributing to the overall reliability of communication in UWA environments.

Traditional approaches such as least square (LS) method is used in literature to determine the channel [4]. In addition, the minimum mean square error (MMSE) estimate is widely

recognized as the optimal solution with respect to minimizing mean-squared error (MSE). But its practical utilization is difficult because of the high complexity of its calculation and requirement of knowledge about channel statistics a priori [5]. On the other hand, DNN have gained great success in many fields [6], [7]. For the case of channel estimation, methods which use neural networks have the ability to learn the mapping from received signal to a channel parameter. However, due to less exploration of their full potential and underlying mechanisms, they do not work efficiently in UWA channel estimation. Conventional deep learning (DL) models are reported to reach lower performance levels in terms of overfitting and generalization on underwater communication systems [6], [7], [8]. More specifically, a significant issue for a critical challenge in developing reliable and efficient models for underwater communication is the scarcity of available underwater channel data to feed into sophisticated DL models [8]. Without real-world data to train on, we are bound for overfitting and failing when applied to new unseen physical data. In this regard, generating adequate amounts of channel conditions is needed to overcome this limitation.

This article presents a DenseNet architecture for channel estimation in UWA OFDM systems. The proposed model is designed to accommodate various scenarios. The DenseNet model is trained using training samples in order to acquire knowledge about the properties of UWA multipath channels. These channels are subject to a range of nonlinear distortions and interferences. We conduct experiments to examine the efficacy of channel estimation and the influence of various modulation schemes, specifically BPSK and QPSK. The performance of the state of the art models including LS, MMSE, FC-NN, and the proposed DenseNet algorithms are evaluated based on the BER versus SNR criterion. The primary findings and contributions of this research are outlined as follows.

- 1) We have introduced the DenseNet estimator used for OFDM channel estimation in UWA communication. This approach, distinguished by its dense connectivity, effectively captures the complexities of the UWA channels. Its inherent robustness even with a limited number of pilots sets it apart from conventional methods. Alongside introducing this novel application, we provide theoretical insights and present a mathematical model illustrating the workings and efficacy of the DenseNet estimator for channel estimation.
- 2) For our training data set, we use a comprehensive collection of UWA channel data using real-time varying CIR extracted from the WATERMARK data set. Specifically, we utilize three WATERMARK channels: BCH, KAU1, and KAU2, and conduct our experiments across a range of SNRs to ensure robustness and versatility in our analysis.
- 3) We undertake rigorous experiments to assess the comparative performances of BPSK and QPSK modulation schemes within the DenseNet-based estimation framework. Furthermore, we benchmark our proposed estimator against conventional channel estimation methods, namely LS, MMSE, and FC-NN, across the three selected watermark channels—BCH, KAU1, and KAU2. The results

unequivocally showcase our approach's superior performance in various aspects, including estimation accuracy, improving BER with fewer pilots, and adaptability to different modulation schemes and channel conditions in challenging UWA environments.

The rest of this article is organized as follows. In Section II, we present a detailed overview of literature. In Section III, the proposed scheme is described in detail. The simulation results are demonstrated in Sections IV. Finally, Section V concludes this article.

II. LITERATURE REVIEW

Estimating the UWA channel is an important step in the underwater communication system and ensures the fidelity and efficacy of transmissions between submerged devices [8]. Research identifies three distinct methodologies for channel estimation. The first one is pilot-assisted schemes, where a pilot signal is sent along with the data subcarriers. Second, blind methods do not use known pilot symbols to estimate the channel. Instead, they use statistical features. Third, semiblind methods find a middle ground between blind methods and pilot-assisted methods by using statistical features and only knowing part of the data being sent. Pilot insertion mainly falls into two categories: 1) comb-type; and 2) block-type configurations. Comb-type pilots are interspersed within the data carriers in the frequency domain, whereas block-type pilots are inserted in specific subcarriers. Block pilots are well-suited for addressing selective fading issues while comb pilots provide a simple and uniform coverage of the frequency domain, making them suitable for channel estimation and synchronization in a broad range of scenarios.

To enhance the efficacy of underwater communication systems, signal processing methods are indispensable. Various techniques, including channel estimation and specifically designed modulation strategies for UWA, are explored by researchers [9]. This section delves into a comprehensive review of literature centered on channel estimation using supervised learning, pilot-assisted approaches, aligning with the direction of our proposed methodology.

DL has revolutionized communication systems, offering significant advances in OFDM communications, particularly when channels present complexities that defy traditional analytical models. The synthesized literature, summarized in Table I, encompasses challenges, research contributions and future directions in channel estimation. The body of work we have examined shows a trend toward using DL to improve performance. Approaches like convolutional neural networks (CNNs) and Long Short Term Memory (LSTM) networks are recurrent themes, indicating their growing prevalence as tools to overcome the nuanced difficulties of UWA channels. For instance, the authors of [10] addressed the issue of multipath channel fading caused by high propagation latency and bottom reflections in UWA OFDM communication. This leads to outdated CSI and inaccurate channel estimation. To solve this problem, the authors proposed the CsiPreNet model which uses a combination of CNN and LSTM. For channel estimation in MIMO OFDM, the authors in [11]

TABLE I
SUMMARY OF RELATED WORK DONE IN UWA CHANNEL ESTIMATION

Problems addressed	Contributions	Techniques	Metrics	Limits/Future
Improving channel estimation in MIMO-OFDM [11]	Developed DL models for enhanced CSI using pilot symbols and CIR	DNN	BER, NMSE	Enhanced layering for improved estimation
Massive MIMO CSI accuracy [12]	Performed scalable DL-based channel estimation for massive MIMO systems	DL	–	Counteracting pilot contamination and external interference
Reducing complexity in MIMO-OFDM channel estimation [13]	Reviewed and evaluated various channel estimation and equalization methods	Comparative Study	Complexity, Performance	Need of low-complexity equalization schemes
Joint channel classification and estimation in UWA [14]	Implemented joint CC, CE, and SD with a DNN, using water type classification for CE/SD optimization	DNN	–	Tailored CE/SD strategies for diverse aquatic conditions
Adaptive denoising for UWA channels [15]	Proposed an adaptive de-noising method using pilot symbols in low SNR UWA conditions	Adaptive Method	–	Expansion to blind/semiblind channel estimation methods
Peak-to-average power ratio and computational cost in UWA OFDM [16]	Introduced Mary spread spectrum with LSTM for UWA communications	LSTM	–	Development of low complexity, efficient interference mitigation schemes
Nonlinear distortion and PAR in UWA OFDM [17]	Utilized a DNN for distortion removal, improving the system performance	DNN	–	Nonlinearity mitigation and model refinement
Simplifying channel estimation process [18-19]	Proposed a DL-based receiver for UWA OFDM systems, avoiding traditional estimation methods	DL Receiver	–	Optimization of the signal processing workflow
Sparse channel estimation in multipath fading [20]	Explored LS sparse channel estimation for multipath fading environments	LS Sparse	BER, MSE	Extension to OFDM-IDMA systems and multiple user scenarios
LS algorithm robustness to noise [21]	Improved LS channel estimation with a weighted averaging approach for pilot symbol modification	Optimized LS	BER, MSE	Evaluation of computational efficiency and dynamic condition adaptability
LS sparse channel estimation optimization [22]	Combined LS estimation with polynomial interpolated optimization for performance enhancement	Optimized LS	BER	Comparative analysis with existing CE techniques, robustness evaluation

emphasized the importance of estimating the CSI. With received pilot symbols and CIR, the proposed DL models were trained for channel estimation. In terms of BER and normalized mean square error, the proposed techniques outperformed the LS and back propagation neural network algorithms and are comparable to the MMSE algorithm. However, when the number of layers are limited, the DNN model is inferior in estimation performance but superior in storage resources and running time.

Within multiantenna frameworks, particularly in MIMO cellular systems, acquiring accurate CSI is pivotal for improving reception tasks. Existing methods have some solutions but also come with challenges. In [12], the authors proposed a new approach utilizing DL to create a proficient and scalable massive MIMO channel estimator. Moreover, the authors in [13] conducted a comprehensive review of various MIMO-OFDM channel estimation methods based on complexity and performance analysis. The authors in [14] proposed a DL-based combined Channel Classification (CC), channel estimation, and Signal Detection (SD) scheme. A DNN-based scheme [30] was proposed recently for channel estimation. Using an offline-trained DNN to classify underwater environments made channel estimation.

An additional innovative adaptive denoising technique for channel estimation applicable to OFDM system was proposed in [15] and its implementation for underwater wireless acoustic communications. This technique uses two sets of pilot symbols: One that serves to estimate the CIR and another that selects the best window for denoising. Proposed method instead of relying on channel statistics or SNR calculates an optimal denoising window. Based on simulation results, it is shown that in the case of estimating a time-varying UWA channel, the proposed method can achieve far better accuracy than a benchmark channel estimators, especially at low SNRs. The authors proposed investigating blind and semiblind channel estimation techniques for MIMO-OFDM to decrease complexity and boost spectral efficiency. In [16], the authors addressed two problems in UWA channel estimation using OFDM: 1) high peak to-average ratio; and 2) high computational cost in despreading and decarrier. To tackle these challenges, they proposed a DL model based on M-ary spread spectrum for UWA communication systems. The model improves the data rate at low signal power levels. They used LSTM at the receiver for direct demodulation. The simulation results and a water tank experiment validated the effectiveness and robustness of the proposed scheme. However, a low complexity and bandwidth-efficient scheme is needed for better interference mitigation. In [17] and [18], a novel approach for the consideration of nonlinear distortion in the case of UWA OFDM communication is addressed. The method they used, employs a DNN to remove these distortions and simultaneously reduce the peak-to-average power ratio. In [19], the authors proposed a DL-based receiver to cope with the challenge of channel estimation in UWA OFDM communication systems. This approach diverges from traditional receivers that require explicit channel estimation and equalization, potentially streamlining the signal processing pipeline.

In contrast to DL models, the authors in [20] presented a simulation study of block-type channel estimation algorithms in multipath fading channels. In addition, the paper introduced a

sparse channel estimation method using LS, showing its superiority for channel estimation. The authors suggested that future research could focus on analyzing the performance of OFDM systems by incorporating multiple users within the LS framework for sparse channel estimation. Furthermore, the authors in [21] utilized the LS for channel estimation. This approach utilizes the channel frequency response of a pilot symbol to adjust subsequent symbols via a weighted averaging technique. In addition, an optimized LS sparse channel estimation algorithm was presented in [22]. They integrated LS estimation with a polynomial interpolated black widow optimization. The proposed method demonstrates superior performance in metrics like BER. However, the paper does not clearly address the computational demands or the robustness under diverse and dynamic underwater conditions, which are crucial for practical implementation. In addition, a more detailed comparison with specific existing channel estimation techniques is needed to fully validate the improvements claimed.

In the preceding discussion, we identified key challenges in UWA communication, including the need for prior channel information, low SNR, and high computational costs. These challenges underscore the necessity for advanced research in real-time, at-sea UWA systems to tackle severe multipath effects and achieve low BER through enhanced channel estimation. Addressing these issues, LS estimation is noted for its simplicity but suffers from noise sensitivity, which is inversely related to SNR [3], [4], [5], [6], [7]. In addition, MMSE estimation offers a solution but requires detailed knowledge of channel and noise variance, presenting challenges underwater [3], [5], [7]. In contrast, FC-NN are recognized for handling complex channel behaviors by capturing nonlinearities, with DNNs trained on realistic models showing promise over traditional methods despite concerns about overfitting, computational demands, and adaptability to variable conditions [21]. This article evaluates these prevalent techniques like LS, MMSE, and FC-NN against our proposed model to address the dynamic challenges in UWA communication.

III. PROPOSED SYSTEM MODEL

The methodology of the proposed system model is explained in the following subsections.

A. UWA-OFDM System Model

A diagram of the UWA-OFDM scheme [23], [24], [25], [26] is depicted in Fig. 1. The binary data sequence is generated and pilot tones are introduced in order to estimate the CIR of the channel model. The data is then mapped using phase shift keying modulation method. The UWA-OFDM system employs the Inverse Fast Fourier Transform (IFFT) to convert the data from frequency to time domain, utilizing N orthogonal narrowband subcarriers. The time domain signal $x(n)$ is derived from the frequency domain signals $X(k)$ by the following equation [11]:

$$x(n) = \text{IFFT}\{X(k)\} = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi \frac{nk}{N}}, \quad n=0, 1, \dots, N-1. \quad (1)$$

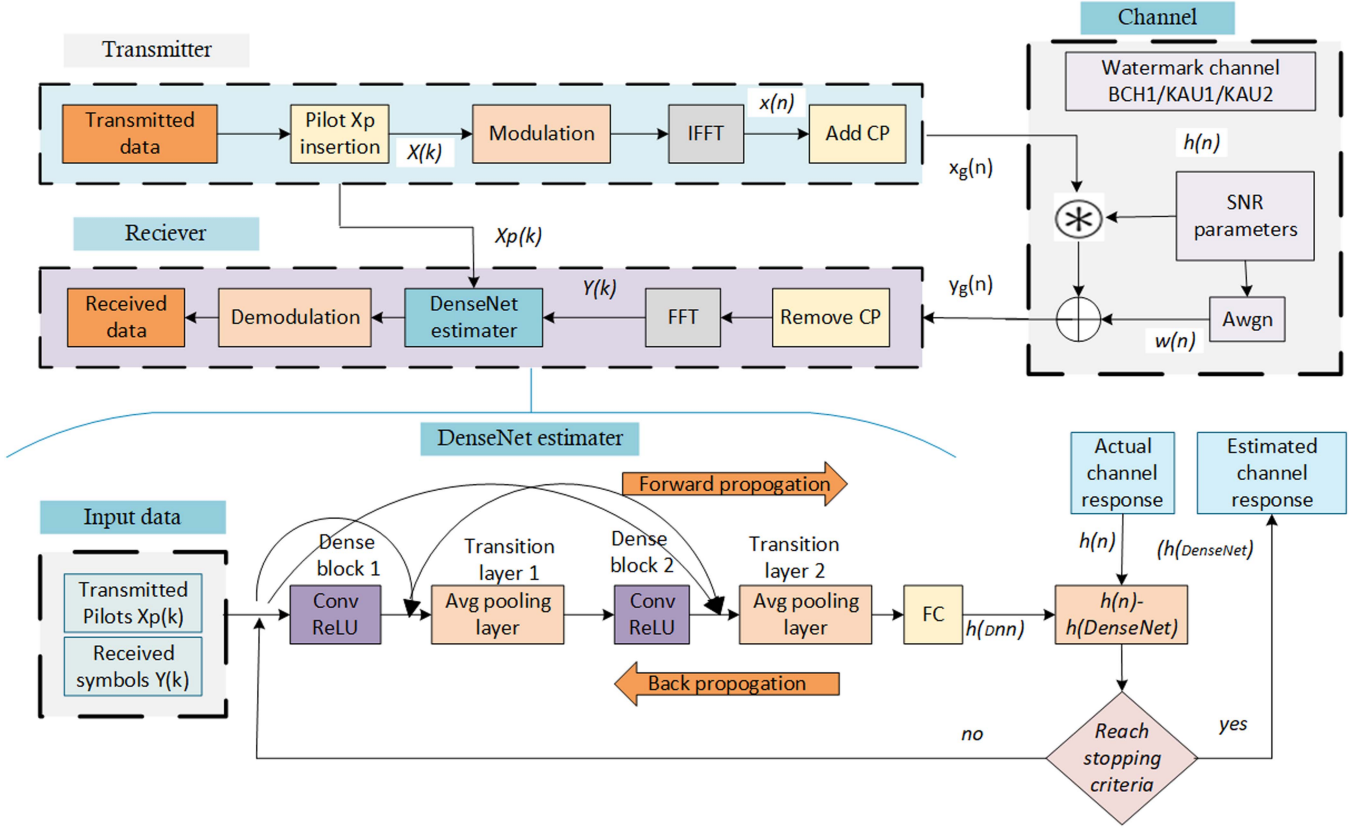


Fig. 1. Architecture of proposed system model.

Following the IFFT operation, the insertion of CP samples serves as guard intervals to mitigate intersymbol interference (ISI). The transmitted signal in the time domain, which incorporates the CP, can be expressed in the following manner [11]:

$$x_g(n) = \begin{cases} x(N+n), & \text{for } n = -N_g, -N_g+1, \dots, -1 \\ x(n), & \text{for } n = 0, 1, \dots, N-1. \end{cases} \quad (2)$$

The variable N_g represents the number of samples in the CP. This process involves duplicating the last N_g samples of the signal $x(n)$ and inserting them as a CP at the beginning of the symbol. As a consequence, the signal $x_g(n)$ is obtained, which has a total length of $N + N_g$.

The received signal $y_g(n)$ is obtained after transmission through the UWA channel [11]

$$y_g(n) = x_g(n) \otimes h(n) + w(n), \quad -N_g \leq n \leq N-1. \quad (3)$$

The operator $\otimes$ denotes circular convolution, whereas $w(n)$ represents Additive White Gaussian Noise (AWGN) with a zero-mean. The CIR, denoted as $h(n)$, can be expressed in the following manner [11]:

$$h(n) = \sum_{i=0}^{r-1} h_i \delta(n - \tau_i). \quad (4)$$

In the given context, δ represents the unit impulse, r denotes the count of multipaths, and h_i and τ_i represent the discrete complex gain and time delay of the i th tap, respectively.

The CP is eliminated in the receiver. The time-domain signal $y(n)$ is converted into the frequency-domain signal $Y(k)$ through the use of Fast Fourier Transform (FFT) operations, according to the following equation [11]:

$$Y(k) = \text{FFT}\{y(n)\} = \frac{1}{N} \sum_{n=0}^{N-1} y(n) e^{-j2\pi \frac{nk}{N}}, \quad k = 0, 1, \dots, N-1. \quad (5)$$

Therefore, if the ISI is reduced fully, the received signal may be expressed as [11]

$$Y(k) = X(k)H(k) + W(k), \quad k = 0, 1, \dots, N-1. \quad (6)$$

The Fourier transforms of $h(n)$ and $w(n)$ are denoted as $H(k)$ and $W(k)$, respectively. The relationship between the sent signal and received signal in an UWA channel can be accurately described using the frequency-domain representations of $H(k)$ and $W(k)$.

B. Proposed DenseNet for UWA Channel Estimation

In this subsection, we outline our DenseNet-based methodology designed for the precise estimation of UWA channels. The DenseNet structure is intricate, incorporating L layers that range

from an input layer, through L-2 hidden layers packed with dense blocks and transition layers, and culminate in an output layer.

1) *DenseNet Network Architecture*: The network architecture of the proposed DenseNet is designed to estimate complex channel gains for individual subcarriers within an OFDM communication system. The architecture consists of the following key layers:

The initial layer of the network serves as the input layer. It receives both the received OFDM symbols $Y(k)$ and the transmitted pilot symbols $Xp(k)$. These symbols capture vital characteristics of the communication channel. In addition, the network architecture comprises two dense blocks with transition layers in between. Within the first dense block, a convolutional layer processes the input data using 3×3 filters, totaling 32 filters. The convolution operation for each filter $k^{(1)}$ is expressed as

$$z_{i,j,k^{(1)}} = \sum_{p=-1}^1 \sum_{q=-1}^1 (x_{i+p,j+q} \cdot w_{p,q,k^{(1)}} + b_{k^{(1)}}). \quad (7)$$

Here, $x_{i,j}$ represents the input data at position (i, j) . $w_{p,q,k^{(1)}}$ denotes the weight for filter $k^{(1)}$ at position (p, q) , and $b_{k^{(1)}}$ is the bias associated with filter $k^{(1)}$. The output of this layer is denoted as $z_{i,j,k^{(1)}}$.

Furthermore, a Rectified Linear Unit (ReLU) layer applies the activation function element-wise to the output of the preceding convolutional layer

$$a_{i,j,k^{(1)}} = \text{ReLU}(z_{i,j,k^{(1)}}) = \max(0, z_{i,j,k^{(1)}}). \quad (8)$$

Following the first dense block, Transition Layer 1 performs average pooling, reducing the spatial dimensions of the data. In this process, each element in the output feature map is computed as the average of a group of elements from the input feature map. The output value $b_{p,q,k^{(1)}}$ at position (p, q) in the pooled feature map is calculated as follows:

$$b_{p,q,k^{(1)}} = \frac{1}{P \cdot Q} \sum_{i=1}^P \sum_{j=1}^Q \left(a((p-1) \cdot S + i, (q-1) \cdot S + j, k^{(1)}) + z_{i,j,k^{(1)}} \right). \quad (9)$$

Here, p and q represent the spatial dimensions of the output feature map, P and Q are the dimensions of the pooling window, and S is the stride. The $k^{(1)}$ index denotes the specific filter or channel.

In addition, the second dense block (Dense Block 2), which follows Transition Layer 1, is structured similarly to Dense Block 1. Within Dense Block 2, a convolutional layer processes the input data using 3×3 filters, totaling 32 filters. The convolution operation for each filter $k^{(2)}$ is expressed as

$$z_{i,j,k^{(2)}} = \sum_{p=-1}^1 \sum_{q=-1}^1 (b_{p,q,k^{(1)}} \cdot w_{p,q,k^{(2)}} + b_{k^{(2)}}). \quad (10)$$

Here, $b_{p,q,k^{(1)}}$ represents the output of Transition Layer 1 at position (p, q) . $w_{p,q,k^{(2)}}$ denotes the weight for filter $k^{(2)}$ at position (p, q) , and $b_{k^{(2)}}$ is the bias associated with filter $k^{(2)}$. The output of this layer is denoted as $z_{i,j,k^{(2)}}$.

Moreover, the output of the previous convolutional layer undergoes to an element-wise application of the activation function by a ReLU layer

$$a_{i,j,k^{(2)}} = \text{ReLU}(z_{i,j,k^{(2)}}) = \max(0, z_{i,j,k^{(2)}}). \quad (11)$$

Following Dense Block 2, Transition Layer 2 performs average pooling. It takes three inputs: The output of Dense Block 2, the output of Dense Block 2, which is concatenated as a skip connection, and the input data concatenated through skip connection. The average pooling process is performed as follows:

$$b_{p,q,k^{(2)}} = \frac{1}{P \cdot Q} \sum_{i=1}^P \sum_{j=1}^Q \left(a((p-1) \cdot S + i, (q-1) \cdot S + j, k^{(2)}) + z_{i,j,k^{(2)}} \right) \quad (12)$$

where: $b_{p,q,k^{(2)}}$ is the output value at position (p, q) in the pooled feature map for filter $k^{(2)}$.

$$a((p-1) \cdot S + i, (q-1) \cdot S + j, k^{(2)})$$

is the element from the output of Dense Block 2. $z_{i,j,k^{(2)}}$ is the element from the input data that was concatenated as a skip connection. P and Q are the dimensions of the pooling window, and S is the stride.

The transition layer downsamples the feature map, reducing its spatial dimensions while preserving important information. This reduction in spatial size helps control the computational complexity of the network while retaining significant features. The Dense Block 2 and transition layer 2 in the architecture enhances the network's ability to capture complex channel characteristics in UWA communication systems. The transition layer downsamples the feature map, reducing its spatial dimensions while preserving important information. This reduction in spatial size helps control the computational complexity of the network while retaining significant features.

The final layer of the network, known as the output layer, is responsible for producing the channel estimation results, denoted as DenseNet. Each output neuron corresponds to a specific subcarrier in the OFDM system, and the network learns to predict the complex channel gain for each subcarrier. The output of this layer is computed as

$$h_{\text{DenseNet}}(k) = \sum_{i=1}^N (w_{k,i} \cdot b_i) \quad (13)$$

where $h_{\text{DenseNet}}(k)$ represents the estimated complex channel gain for subcarrier k . N is the total number of neurons (output units) in the fully connected (FC) layer, $w_{k,i}$ is the weight associated with the connection from neuron i to the output for subcarrier k . b_i is the output value of neuron i in the FC layer. The network's output $h_{\text{DenseNet}}(k)$ for each subcarrier provides the channel estimation result for the corresponding subcarrier.

2) *Training Process*: The network harnesses a supervised learning paradigm for its training process. This involves the utilization of paired data, where each pair is composed of an input encompassing both received OFDM symbols and transmitted pilot symbols and a corresponding output, which encapsulates the

Algorithm 1: DenseNet for UWA Channel Estimation.

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1: Initialize the DenseNet: input layer, hidden layers, and
   output layer.
2: OFDM symbols  $Y(k)$  and pilot symbols  $X_p(k)$  are
   provided at the input layer.
3: Add convolutional layer: 32 filters, size  $3 \times 3$ .
4: for each filter  $k^{(1)}$  and position  $(i, j)$  in input do
5:   Compute convolution  $z_{i,j,k^{(1)}}$  using Eq. (7).
6:   Apply ReLU to get  $a_{i,j,k^{(1)}}$  using Eq. (8).
7: end for
8: Add average pooling: with window size  $p * q$  and stride
    $S$ .
9: for each filter  $k^{(1)}$  and position  $(p, q)$  do
10:  Compute pooled value  $b_{p,q,k^{(1)}}$  using Eq. (9).
11: end for
12: Add convolutional layer: 32 filters, size  $3 \times 3$ .
13: for each filter  $k^{(2)}$  and position  $(i, j)$  do
14:  Compute convolution  $z_{i,j,k^{(2)}}$  using Eq. (10).
15: Apply ReLU to get  $a_{i,j,k^{(2)}}$  using Eq. (11).
16: end for
17: Add average pooling: with window size  $p * q$  and stride
    $S$ .
18: for each filter  $k^{(2)}$  and position  $(p, q)$  do
19:  Compute pooled value  $b_{p,q,k^{(2)}}$  using Eq. (12).
20: end for
21: Output layer produces channel gains  $h_{\text{DenseNet}}$ .
22: for each OFDM subcarrier  $k$  do
23:  Compute channel estimation  $h_{\text{DenseNet}}(k)$  using Eq.
     (13).
24: end for

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estimated channel characteristics. Through iterative refinement, the network fine-tunes its internal parameters, namely weights and biases, to enhance its proficiency at correlating the input data with precise channel gain values. The operational mechanics of the DenseNet framework are visually represented in Fig. 1 and concisely encapsulated in Algorithm 1 for a clear and systematic understanding.

3) *Data Generation and Training Options:* In the simulation phase, training and testing data sets are constructed in such a way that there would be sufficient data to be used for model training and testing with a minimum risk of overfitting and maximum robustness. We use three channels: BCH, KAU1, and KAU2. Each data set has been generated under different channel conditions. Specifically for the BCH channel data set, which is created by conducting the simulation, contains 5820 samples during the simulation phase. The shape of the input for each sample of the received signal for training is $(N_t, N_{fft}, \text{num})$ that is $582 \times 4096 \times 10$. N_t is the number of samples, N_{fft} is the received signal as input features, and num is the number of iterations. Thus, the BCH data set encompasses 23 838 720 entries in total. Besides, KAU1 and KAU2 channels are produced by the simulation with the input shape of received signal for training is $258 \times 4096 \times 10$, hence, there are 10 557 680 entries in each data set KAU1 and KAU2. To

TABLE II
MODEL TRAINING OPTIONS FOR DENSENET

Parameter	Value
Optimization Algorithm	Adam
Max Epochs	120
Activation	ReLU
Mini batch Size	128
Learning Rate	1e-3
Validation Frequency	30
Shuffle	“every-epoch”
Validation Patience	5

ensure reliable evaluation, we split each data set into 75% for training and 25% for testing, ensuring enough samples in both sets to assess model performance and prevent overfitting. This approach helped maintain the generalization capability of the model while mitigating the risk of overfitting. In addition, training options are pivotal in shaping the effectiveness of the model’s learning phase. Table II provides a comprehensive overview of these options for the DenseNet architecture and explains how they impact the learning process. Adam Optimization algorithm is used to train the neurons since for its adaptive learning capabilities and that enhances the speed of convergence and overall performance of the model especially in cases of noisy gradient. The way Adam can vary the learning rate for each parameter prevents problems due to insufficient convergence and poor stability during training. Since the model is trained for a number of epochs, we set it at value of 120 to make sure that the model runs as fast as possible through all of those iterations before overfitting. In addition, the ReLU activation function is nonlinear and allows for learning of complex patterns and fast convergence due to solving the vanishing gradient problem. Minibatch size of 128 offers a compromise between minibatch practicalities and good constant gradient estimates providing good training stability efficiency. In addition, the learning rate of 0.001 manages step sizes during parameter updates. The model is assessed every 30 epochs through validation, which provides regular updates of accuracy without an excessive slowdown in training. A validation patience of 5 epochs is set to stop training if no improvement in validation performance is observed, preventing overfitting and saving computational resources. Combined these training options provide a systematic and efficient training process, hitting the right balance between convergence speed and accuracy, and thus minimizing overfitting on the DenseNet model while maximizing its performance in general.

After training, the DenseNet-based channel estimator can predict channel gains well so that it can effectively learn UWA channels with a high fidelity. Our methodology is evaluated in detail using key performance measures, specifically the BEB, tested across a wide range of SNRs and number of pilot. The experiments reveal the estimator’s proficiency, revealing its adeptness at minimizing pilot signal overhead while maintaining communication reliability, even in the notoriously difficult underwater conditions. This DenseNet-based UWA channel estimation representative has an impressive ability for accurate prediction of channel gains. This model demonstrates strong robustness

to variable channel conditions in underwater communications, which in conjunction with its ability to significantly reduce pilot overhead makes it a better choice for enhancing the reliability of underwater transmissions.

C. Channel Equalization for OFDM Systems Using DenseNet-Based Estimates

The objective of channel equalization is to mitigate the adverse effects of the communication channel, ensuring the integrity of the transmitted signal. After obtaining the channel's frequency response estimates through the DenseNet, the LS equalization method is employed in the frequency domain to counteract the distortion introduced by the channel. The equalization formula for the k th subcarrier can be represented as

$$Y_{eq}(k) = \frac{Y(k)}{H_{DenseNet}(k)} \quad (14)$$

where

- 1) $Y(k)$ denotes the received signal in the frequency domain for the k th subcarrier.
- 2) $H_{DenseNet}(k)$ is the DNN's estimated channel response for the k th subcarrier.
- 3) $Y_{eq}(k)$ symbolizes the equalized signal in the frequency domain, which is the signal postcompensation for the k th subcarrier.

The incorporation of DenseNet-based estimates provides a robust mechanism to adapt to varying channel conditions, enhancing the reliability and performance of the OFDM system. The LS equalization technique, leveraging these estimates, ensures that the retrieved signal closely resembles the originally transmitted signal, achieving effective communication.

IV. SIMULATION RESULTS AND CRITICAL ANALYSIS

This section begins with an introduction to the experimental environment, utilizing the WATERMARK data set [27], as detailed in Section IV-A. In Section IV-B, we describe how the benchmark schemes. In Section IV-C, the performance metrics is discussed. Section IV-D is concerned with experimental results and their analysis.

The proposed models have been implemented in MATLAB, and the simulation incorporates a set of key parameters listed in Table III. This succinct table offers a comprehensive overview of vital parameters, including modulation strategy, the number of subcarriers, pilot arrangement, FFT dimensions, and the utilization of a CP as a guard interval. Comb pilot insertion is chosen for its spectral efficiency and simple channel estimation process. It allows for consistent pilot spacing across the frequency band, facilitating easier interpolation of the channel response and effective compensation for Doppler shifts, which are common in UWA communications. In addition to these, the Table III also outlines critical system-specific details such as the frequency bands used for OFDM signal, their respective subcarrier spacing, and coherence bandwidths. Notably, the bandwidths of OFDM signal during simulations were in line with the channel bandwidths: 5 kHz for the BCH channel and 4 kHz for the KAU channel. The subcarrier spacing for both

TABLE III
OVERVIEW OF SYSTEM PARAMETERS FOR UWA-OFDM

Parameters	Value
UWA modulation scheme	OFDM
Subcarriers	1024
Pilots	256
Pilot insertion	Comb
FFT size	4096
Guard interval	CP
CP size	256
OFDM signal Bandwidth (BCH)	32.5–37.5 kHz
OFDM signal Bandwidth (KAU)	4–8 kHz
Subcarrier spacing (BCH)	4.88 Hz
Subcarrier spacing (KAU)	3.90 Hz
Coherence bandwidth (BCH)	9.78 Hz
Coherence bandwidth (KAU)	7.81 Hz
Sampling frequency	48 k
Noise model	AWGN
SNR	20 dB/-10:5:20 dB

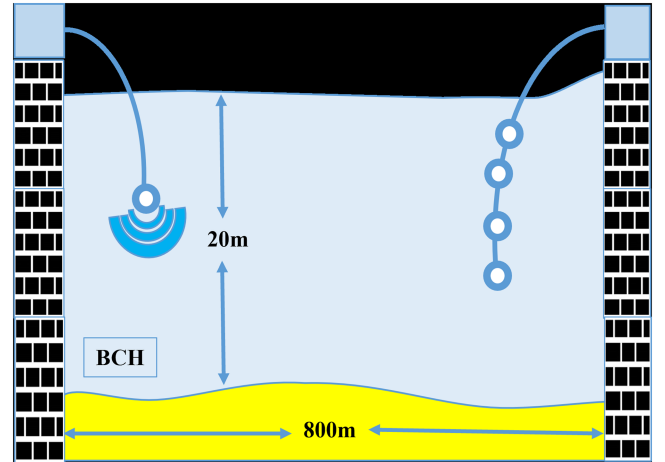


Fig. 2. Schematic diagram of BCH channel.

channels are 4.88 Hz for the BCH channel and 3.90 Hz for the KAU channel. The coherence bandwidth for BCH is 9.78 and for KAU is 7.81, calculated using the formula from [28]. For both the channels, the subcarrier spacing was smaller than the coherence bandwidth, allowing the system to exploit the frequency diversity of the channel. The assumed sampling frequency for these simulations was 48 kHz, ensuring accurate signal representation and processing within the specified frequency bands.

A. Experimentation Environment Based on WATERMARK

For experimentation, we employed UWA channels from the WATERMARK data set, which were obtained from underwater measurements. The transmitter was placed in the water column at BCH (Brest Commercial Harbour) and towed by a surface ship at KAU1 and KAU2 (Kauai) throughout these experiments. The receiver, depicted in Figs. 2 and 3, comprises a vertically suspended hydrophone array. Table IV conducts a comparative

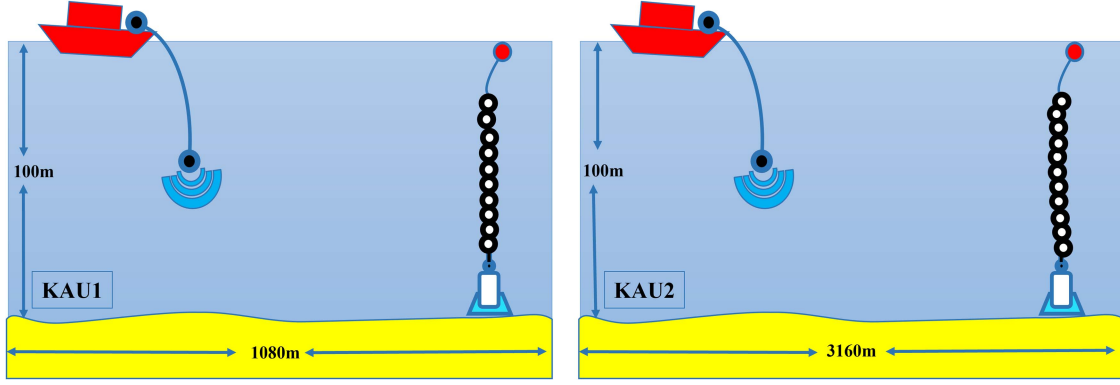


Fig. 3. Schematic diagram of KAU1 and KAU2 channels.

TABLE IV
EXPERIMENTATION SETUP OF DIFFERENT WATERMARK CHANNELS

Name	BCH	KAU1	KAU2
Environment	Harbour	Shelf	Shelf
Time of year	June	July	July
Range	800 m	1080 m	3160 m
Water depth	20 m	100 m	100 m
Transmitter depl.	Suspended	Towed	Towed
Receiver depl.	Suspended	Suspended	Suspended
Frequency range	32.5–37.5 kHz	4–8 kHz	4–8 kHz
Sounding duration	59.4 s	32.9 s	32.9 s
Delay coverage	102 ms	128 ms	128 ms
Doppler coverage	9.8 Hz	7.8 Hz	7.8 Hz
Type	SIMO	SIMO	SIMO
Element spacing	1 m	3.75 m	3.75 m
# cycles	1	1	1
Total play time	1 min	33 s	33 s

analysis of three distinct UWA environments: BCH, KAU1, and KAU2. Notably, BCH is situated in a Harbour, while KAU are located in shelf environments as shown in Figs. 2 and 3, respectively, shows variations in transmission range and water depth. In addition, KAU1 and KAU2 operate in a lower frequency range (4–8 kHz) in contrast to BCH (32.5–37.5 kHz). In addition, there are differences in the delay coverage, with BCH offering a longer duration of 102 ms. With help of this knowledge, we are able to choose the optimum experiment parameters.

B. System and Methods

The benchmark models LS, MMSE, and FC-NN estimator offer different approaches to UWA channel estimation, each with its own strengths and limitations. These state-of-the-art methods are explained and analyzed in the following subsections as they are used as benchmark schemes for comparison.

1) *Least Square*: The LS algorithm is often used for channel estimation. This method aims to minimize the square of the error between received pilots and transmitted pilots. The LS estimation is used to minimize the following cost function [29]:

$$J(\hat{\mathbf{H}}) = \|\mathbf{Y} - \mathbf{X}\hat{\mathbf{H}}\|^2 \quad (15)$$

where $\mathbf{Y}$ is the received signal vector, $\mathbf{X}$ represents the known transmitted training symbols, and $\hat{\mathbf{H}}$ is the estimated channel

vector. This cost function can be expanded in [29] as

$$\begin{aligned} J(\hat{\mathbf{H}}) &= (\mathbf{Y} - \mathbf{X}\hat{\mathbf{H}})^H (\mathbf{Y} - \mathbf{X}\hat{\mathbf{H}}) \\ &= \mathbf{Y}^H \mathbf{Y} - \mathbf{Y}^H \mathbf{X}\hat{\mathbf{H}} - \hat{\mathbf{H}}^H \mathbf{X}^H \mathbf{Y} + \hat{\mathbf{H}}^H \mathbf{X}^H \mathbf{X}\hat{\mathbf{H}}. \end{aligned} \quad (16)$$

To minimize this cost function, we take the derivative of $J(\hat{\mathbf{H}})$ with respect to $\hat{\mathbf{H}}$ and set it equal to zero

$$\frac{\partial J(\hat{\mathbf{H}})}{\partial \hat{\mathbf{H}}} = -2(\mathbf{X}^H \mathbf{Y})^* + 2(\mathbf{X}^H \mathbf{X}\hat{\mathbf{H}})^* = 0. \quad (18)$$

Solving this equation gives the LS channel estimate $\hat{\mathbf{H}}_{\text{LS}}$ as [29]

$$\hat{\mathbf{H}}_{\text{LS}} = (\mathbf{X}^H \mathbf{X})^{-1} \mathbf{X}^H \mathbf{Y} = \mathbf{X}^{-1} \mathbf{Y}. \quad (19)$$

Here, $\mathbf{X}^H$ represents the Hermitian (conjugate transpose) of the transmitted symbol matrix $\mathbf{X}$, $(\mathbf{X}^H \mathbf{X})^{-1}$ is the inverse of $\mathbf{X}^H \mathbf{X}$.

Let's denote each component of the LS channel estimate $\hat{\mathbf{H}}_{\text{LS}}$ by $\hat{H}_{\text{LS}}[k]$, where $k = 0, 1, 2, \dots, N-1$. Since $\mathbf{X}$ is diagonal due to the ICI-free condition, the LS channel estimate $\hat{\mathbf{H}}_{\text{LS}}$ for each subcarrier is

$$\hat{H}_{\text{LS}}[k] = \frac{Y[k]}{X[k]^*}, \quad k = 0, 1, 2, \dots, N-1 \quad (20)$$

where $Y[k]$ is the received signal for subcarrier k , $X[k]$ is the transmitted signal for subcarrier k .

The MSE of this LS channel estimate is given in [29] as

$$\text{MSE}_{\text{LS}} = \mathbb{E} \left\{ (\mathbf{H} - \hat{\mathbf{H}}_{\text{LS}})^H (\mathbf{H} - \hat{\mathbf{H}}_{\text{LS}}) \right\} \quad (21)$$

where $\mathbf{H}$ is the actual channel matrix.

Expanding this expression

$$= \mathbb{E} \left\{ (\mathbf{H} - \mathbf{X}^{-1} \mathbf{Y})^H (\mathbf{H} - \mathbf{X}^{-1} \mathbf{Y}) \right\} \quad (22)$$

$$= \mathbb{E} \left\{ (\mathbf{X}^{-1} \mathbf{Z})^H (\mathbf{X}^{-1} \mathbf{Z}) \right\} \quad (23)$$

where $\mathbf{Z}$ is the noise vector.

$$= \mathbb{E} \left\{ \mathbf{Z}^H (\mathbf{X} \mathbf{X}^H)^{-1} \mathbf{Z} \right\} \quad (24)$$

$$= \frac{\sigma_Z^2}{\sigma_X^2} \quad (25)$$

where σ_Z^2 is the noise variance and σ_X^2 is the variance of the transmitted symbols.

2) *MMSE Estimator*: The MMSE estimation is proposed as a solution to the noise-prone flaws of LS estimation. This method aims to reduce the MSE between the actual channel and its estimation. However, obtaining the auto correlation function of the channel and the noise variance, which is necessary for this task, can be challenging in an underwater environment. It can also be computationally expensive, making it inappropriate for real-time implementation in underwater communication systems, however, it provides enhanced performance compared to LS estimation. Mathematically, it can be expressed in [29] as follows.

Consider the LS solution in (19), $\hat{\mathbf{H}}_{LS} = \mathbf{X}^{-1}\mathbf{Y} \triangleq \hat{\mathbf{H}}$. By using the weighting matrix $\mathbf{W}$, we define the MMSE estimate as $\hat{\mathbf{H}} \triangleq \mathbf{W}\tilde{\mathbf{H}}$. The MSE for the channel estimate $\hat{\mathbf{H}}$ is expressed as

$$J(\hat{\mathbf{H}}) = \mathbb{E}\{||\mathbf{e}||^2\} = \mathbb{E}\{||\mathbf{H} - \hat{\mathbf{H}}||^2\}. \quad (26)$$

The MMSE channel estimation technique seeks to optimize the linear estimator (i.e., $\mathbf{W}$) to minimize the MSE in (26). According to the orthogonality principle, the estimation error $\mathbf{e} = \mathbf{H} - \hat{\mathbf{H}}$ must be orthogonal to the observed channel estimate $\tilde{\mathbf{H}}$, which gives us the following condition:

$$\begin{aligned} \mathbb{E}\{\mathbf{e}\tilde{\mathbf{H}}^H\} &= \mathbb{E}\{(\mathbf{H} - \hat{\mathbf{H}})\tilde{\mathbf{H}}^H\} \\ &= \mathbb{E}\{(\mathbf{H} - \mathbf{W}\tilde{\mathbf{H}})\tilde{\mathbf{H}}^H\} \\ &= \mathbb{E}\{\mathbf{H}\tilde{\mathbf{H}}^H\} - \mathbf{W}\mathbb{E}\{\tilde{\mathbf{H}}\tilde{\mathbf{H}}^H\} \\ &= \mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}} - \mathbf{W}\mathbf{R}_{\tilde{\mathbf{H}}\tilde{\mathbf{H}}} = 0 \end{aligned} \quad (27)$$

where $\mathbf{R}_{\mathbf{AB}}$ denotes the cross-correlation matrix of two matrices $\mathbf{A}$ and $\mathbf{B}$, defined as $\mathbf{R}_{\mathbf{AB}} = \mathbb{E}\{\mathbf{AB}^H\}$. The LS channel estimate $\hat{\mathbf{H}}$ is given by [29]

$$\hat{\mathbf{H}} = \mathbf{X}^{-1}\mathbf{Y} = \mathbf{H} + \mathbf{X}^{-1}\mathbf{Z}. \quad (28)$$

Solving for the weighting matrix $\mathbf{W}$ from (28) yields

$$\mathbf{W} = \mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}}\mathbf{R}_{\tilde{\mathbf{H}}\tilde{\mathbf{H}}}^{-1}. \quad (29)$$

The matrix $\mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}}$, which is the autocorrelation matrix of $\tilde{\mathbf{H}}$, can be expressed as [29]

$$\begin{aligned} \mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}} &= \mathbb{E}\{\tilde{\mathbf{H}}\tilde{\mathbf{H}}^H\} \\ &= \mathbb{E}\{(\mathbf{X}^{-1}\mathbf{Y})(\mathbf{X}^{-1}\mathbf{Y})^H\} \\ &= \mathbb{E}\{(\mathbf{H} + \mathbf{X}^{-1}\mathbf{Z})(\mathbf{H} + \mathbf{X}^{-1}\mathbf{Z})^H\} \\ &= \mathbb{E}\{\mathbf{H}\mathbf{H}^H\} + \mathbb{E}\{\mathbf{X}^{-1}\mathbf{Z}\mathbf{H}^H\} \\ &\quad + \mathbb{E}\{\mathbf{H}\mathbf{Z}^H(\mathbf{X}^{-1})^H\} + \mathbb{E}\{\mathbf{X}^{-1}\mathbf{Z}\mathbf{Z}^H(\mathbf{X}^{-1})^H\} \\ &= \mathbf{R}_{\mathbf{HH}} + \frac{\sigma_Z^2}{\sigma_X^2}\mathbf{I}. \end{aligned} \quad (30)$$

Here, $\mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}}$ represents the cross-correlation matrix between the true channel vector and the temporary channel estimate vector

in the frequency domain. By using (27), the MMSE channel estimate can be expressed in [29] as

$$\hat{\mathbf{H}} = \mathbf{W}\tilde{\mathbf{H}} = \mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}}\mathbf{R}_{\tilde{\mathbf{H}}\tilde{\mathbf{H}}}^{-1}\tilde{\mathbf{H}} = \mathbf{R}_{\mathbf{H}\tilde{\mathbf{H}}}\left(\mathbf{R}_{\mathbf{HH}} + \frac{\sigma_Z^2}{\sigma_X^2}\mathbf{I}\right)^{-1}\tilde{\mathbf{H}}. \quad (31)$$

3) *FC-NN*: The DL methods used in UWA channel estimation is based on the backpropagation neural network [30]. It capture complex nonlinear relationships by mapping the CIR with received signal and the transmitted pilots. The FC-NN used in this article comprises three dense layers. There are 64 nodes in the first dense layer, and to reduce overfitting, there is a dropout layer with a 0.1 dropout rate. The second dense layer has 1024 nodes, followed by a ReLu layer. This configuration allows the NN to effectively learn and represent the underlying mapping between input features and the CIR in the UWA channel.

The benchmark model FC-NN, MMSE estimator, and LS have unique method characteristics for UWA channel estimation. For cases where there is a very complex behaviour of the underlying channel, the FC-NN can learn highly nonlinear relations. However, it is prone to overfitting if the model complexity is not properly controlled.

C. Performance Evaluation Metrics

There are many evaluation metrics that fall under UWA communication. These are standard measures such as MSE, which play an important role in evaluating DL models. Moreover, it is common practice to show BER versus SNR and also BER versus pilots for communication quality which provides useful information about the system performance.

MSE measures the average squared difference between predicted and actual values, indicating the magnitude of prediction errors and is frequently employed in regression tasks and can be utilized to evaluate the precision of predictions in UWA communication. However, its sensitivity to outliers can be a drawback because the squaring operation accentuates larger errors. The MSE is computed as [31]

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (32)$$

where n represents the total samples, y_i denotes the actual value of the target variable for the i th sample, and $\hat{y}_i$ represents the predicted value of the target variable for the i th sample.

Conversely, R2 score [31] measures the goodness of fit of the regression model with respect to data. The higher R2 score and Explained Variance, the better fit of model to data, and the more dependence of dependent acoustic signal on independent variables. Yet, it must be the analysis of these metrics which can lead to a pitfall. R2 score and Explained Variance can be influenced by irrelevant variables and overfitting, which may artificially inflate their values. Therefore, it is crucial to carefully select relevant variables and assess the model's generality to ensure the validity of these metrics in the UWA communication

domain. The R2 score is calculated in [31] as

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (33)$$

where $\bar{y}$ is mean value of the target variable over all samples

SNR and BER are important parameters in signal processing for UWA communication. SNR is the ratio of power of a signal to its background noise; it serves as an indicator of signal quality and how immune the signal is from disruption by noise. Higher SNR values indicate stronger and more reliable signals, while lower SNR values suggest susceptibility to noise [3]. Evaluating SNR helps assess signal transmission accuracy in challenging underwater environments. BER quantifies the ratio of bits received in error to total transmitted bits. It directly measures how accurate data transmits. Analyzing BER is crucial for assessing the system's performance in terms of data accuracy and reliability.

D. Experimental Results

The experimental results are explained in the following subsections.

1) *DenseNet Generalized Performance in Channel Estimation:* DenseNet includes dense blocks in which each layer receives inputs from all preceding layers. This breaks the isolation between layers, allowing for efficient feature reuse and better gradient flow during backpropagation. The dense connectivity ensures that gradients do not vanish or explode, leading to more stable and effective training. By providing shorter paths for information flow and gradient propagation, DenseNet mitigates the vanishing gradient problem. The proposed DenseNet effectively learns feature maps representing the frequency, temporal, and noise characteristics of underwater acoustic signals. By training on paired data consisting of transmitted OFDM symbols (pilots) $X_p(k)$ and received symbols $Y(k)$, the network can learn the mapping between received signal and the CIR, which is essential for accurate channel estimation. Visualizing these feature maps allows us to observe how the network distinguishes between different signal components, guided by actual channel responses for various environments such as BCH (Harbor), KAU1, and KAU2 (Shelf). DenseNet uses a hierarchical learning procedure in which early layers represent low-level signal properties, and deep layers represent higher level patterns and dependencies. Such, ensures the network to learn both the local and global features of the underwater acoustic environment, enhancing its capacity in learning correct mappings and improving generalization ability leading to more accurate and robust channel estimation. In the experiments conducted, on channels BCH, KAU1, and KAU2, the channels in deployment phase were produced from UWA environment which used during training phase. The most important thing for this, to ensure that the model will work and be robust in changing environmental conditions, is to know how does the response of the model behave when using it with discrepancies. These three environments then use a DenseNet model trained using the same architecture and complexity. This consistent training approach confirms the DenseNet model's robustness, as it consistently

produces enhanced channel estimates. The results depicted in the accompanying figures substantiate this, thereby validating the model's general applicability across varying conditions.

In addition, the dense connectivity in DenseNet, where each layer receives inputs from all preceding layers, facilitates efficient feature reuse and enhances gradient flow during back-propagation. The dense connectivity resulting in more stable and effective training. In our proposed scheme, the training samples of the received signal, along with the transmitted pilots, are incorporated into the DenseNet model. After conducting several iterations of training, we have acquired the MSE loss and the number of training rounds, as depicted in Fig. 4. It can be observed that the training loss curve exhibits a decreasing trend as the number of epochs increases, eventually reaching its minimum value. To mitigate overfitting, the training rounds were adjusted to 300 for KAU and 120 for Channel BCH in subsequent analysis, as shown in Fig. 4(a)–(c). The MSE loss of the DenseNet estimator for the real and imaginary parts is 0.1017 for BCH, 0.0902 for KAU1, and 0.0901 for KAU2 channel.

Furthermore, the proposed model has the ability to reuse features learned from earlier layers, which is crucial for capturing intricate patterns in underwater acoustic signals that are often affected by noise and multipath propagation. This feature reuse mechanism allows our model to build upon previously learned representations, enhancing its ability to model complex acoustic environments. As a result, the proposed DenseNet estimator shows reliable performance in TVIR estimation, as illustrated in Fig. 5. The findings indicate that the TVIR estimation using DenseNet for UWA-OFDM closely approximates the true TVIR. Simultaneously, DenseNet demonstrates lower amplitude and phase errors compared to other methods, improving channel estimation accuracy. Dense connections enable DenseNet to use features, which is proposed to achieve more accurate amplitude and phase estimations. DenseNet outperforms other channel estimators in terms of amplitude and phase error, which allows us to achieve stable performance for UWA channel estimation. LS estimator performs much worse than DenseNet in terms of estimation, as illustrated in Fig. 6. It fails to model the large dynamic range and nonlinearity of UWA channels as its channel estimation inaccuracies are much larger. The MMSE estimator outperforms LS, but DenseNet does much better than MMSE with little improvement. The results of amplitude and phase error across different subcarriers, as depicted in Fig. 6(a)–(c) further corroborate these findings. Furthermore, DenseNet utilizes fewer parameters than most classical convolutional networks with the same depth to cause less overfitting and increase the computational efficiency. As from the R-squared scores of 0.9701 for BCH, 0.9100 for KAU1 and 0.8610 for KAU2, we can see that DenseNet generalizes well along different channels. In addition to leading DenseNet to being robust against noise, the parameter efficiency of this architecture makes it effective even when trained on diverse and challenging underwater acoustic domains. The model's remarkable capability of reducing both amplitude and phase errors demonstrates its promise as an effective UWA channel estimation solution.

Two dense blocks offers a compromise between model complexity and computational cost. Though adding more dense

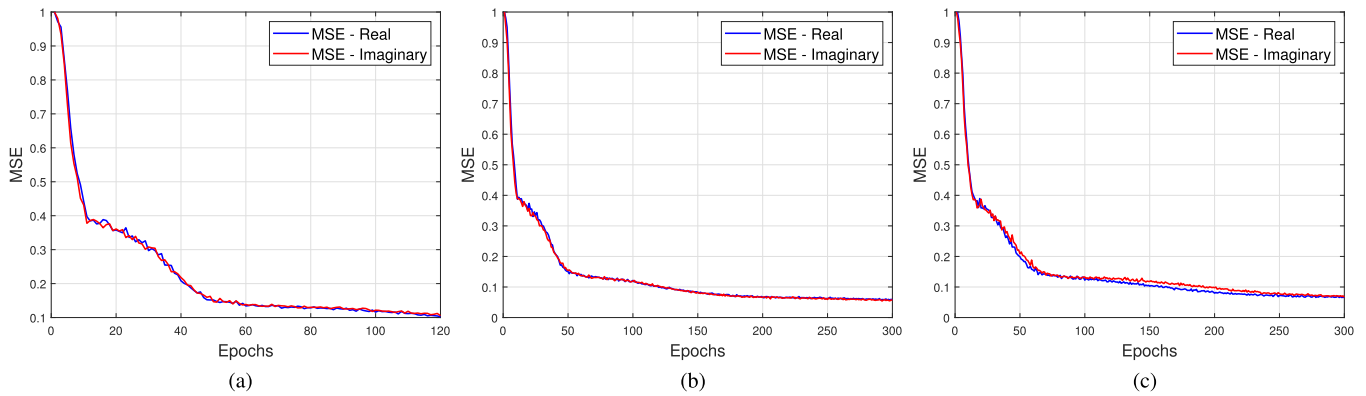


Fig. 4. MSE of DenseNet for BCH and KAU channels. (a) The MSE loss of DenseNet on BCH channel. (b) The MSE loss of DenseNet on KAU1 channel. (c) The MSE loss of DenseNet on KAU2 channel.

blocks could increase performance by capturing more complex patterns, they also add computational costs and risk overfitting. In particular, the convolutional layers with ReLU activation and transition layers with average pooling in DenseNet's architecture are important. Convolutional layers serve to dig out local patterns in the data, while ReLU act as the nonlinear factor, giving an ability for complex pattern learning to our model. Because transition layers decrease the spatial dimension of feature maps, balancing model complexity and overfitting is possible. The FC layer at the end combines features extracted by dense blocks, performing the final channel estimation. This architectural setup allows DenseNet to achieve explained variance scores of 0.9200 for BCH, 0.9101 for KAU1, and 0.8911 for KAU2, indicating its robust performance.

In addition, compared to the other methods (LS, MMSE, FC-NN), DenseNet achieves lower BER values in all channels and for both BPSK and QPSK modulation schemes. The better BER performance implies that DenseNet is able to learn and generalize on channel representations even in poor conditions. This indicates that DenseNet reduces the BER by nearly 20% relative to FC-NN showing its robustness in channel estimation. Unlike traditional estimators (LS and MMSE), the FC-NN estimator presents performance closely rivaling DenseNet. Its error rates, though not as minimal as those of DenseNet, are significantly better than those of LS and MMSE, positioning it as a viable contender for channel estimation under UWA conditions.

In addition, DenseNet's ability to handle the intricacies of underwater acoustic environments is further evidenced by its lower BER and MSE values compared to other estimators. DenseNet captures both local and global features thanks to its dense connectivity and architectural, resulting in more accurate and reliable channel estimation. This becomes especially useful in underwater acoustic communication where environmental conditions have a wide variance. These results suggest that the performance consistency of DenseNet over varying channels and modulation schemes makes it an appealing starting point for a general purpose approach to UWA channel estimation, outperforming competitive techniques such as LS and MMSE, but also more advanced approaches like FC-NN.

2) Impact of Modulation Scheme: In order to provide a more comprehensive understanding of the impact of various modulation schemes on the efficacy of these channel estimation methods, the BER performance of the LS, FC-NN, DenseNet, and MMSE methods is depicted in Fig. 7. When examining the performance presented in Fig. 7, it becomes evident that across both BCH and KAU, the values of BER for the four methods under consideration exhibit a decreasing trend as the SNR increases from -10 to 20 dB. This observation holds true for both the BPSK and QPSK modulation schemes. This phenomenon can be attributed to the inverse relationship between the SNR and the noise variance in underwater acoustic multipath channels. As the SNR increases, the noise power decreases, consequently leading to a reduction in the noise variance. In the context of our study, we have conducted a comparative analysis between our proposed DenseNet model and three other methods, namely LS, FC-NN, and MMSE approaches. Specifically, we have evaluated the BER performance of these methods for two different modulations, i.e., BPSK and QPSK. Our findings indicate that for all values of SNRs, the BER of BPSK modulation is better than QPSK modulation. This implies that BPSK exhibits superior performance compared to QPSK in our experimental setup.

To test the effect of modulation schemes, for instance, in Fig. 7(a), at an SNR of 20 dB, the BER with BPSK is roughly 0.0004 , whereas in Fig. 7(d), the values with QPSK are 0.0009 . Because increasing modulation order improves the requirement for neural network learning accuracy, compared to a channel estimator with a lower modulation order, a higher modulation order finds it more challenging to correctly categorize the received signal to distinct constellation points. Indeed, the expected findings show the logic and validity of our suggested DenseNet technique.

3) BER Analysis of Proposed Scheme: Based on the findings depicted in Fig. 7 for each channel, it can be observed that the BER values for all the aforementioned methodologies exhibit a gradual decline as the SNR escalates from -10 to 20 dB. Among the estimators considered, the LS algorithm exhibits the poorest performance across the three channels. In contrast, the MMSE

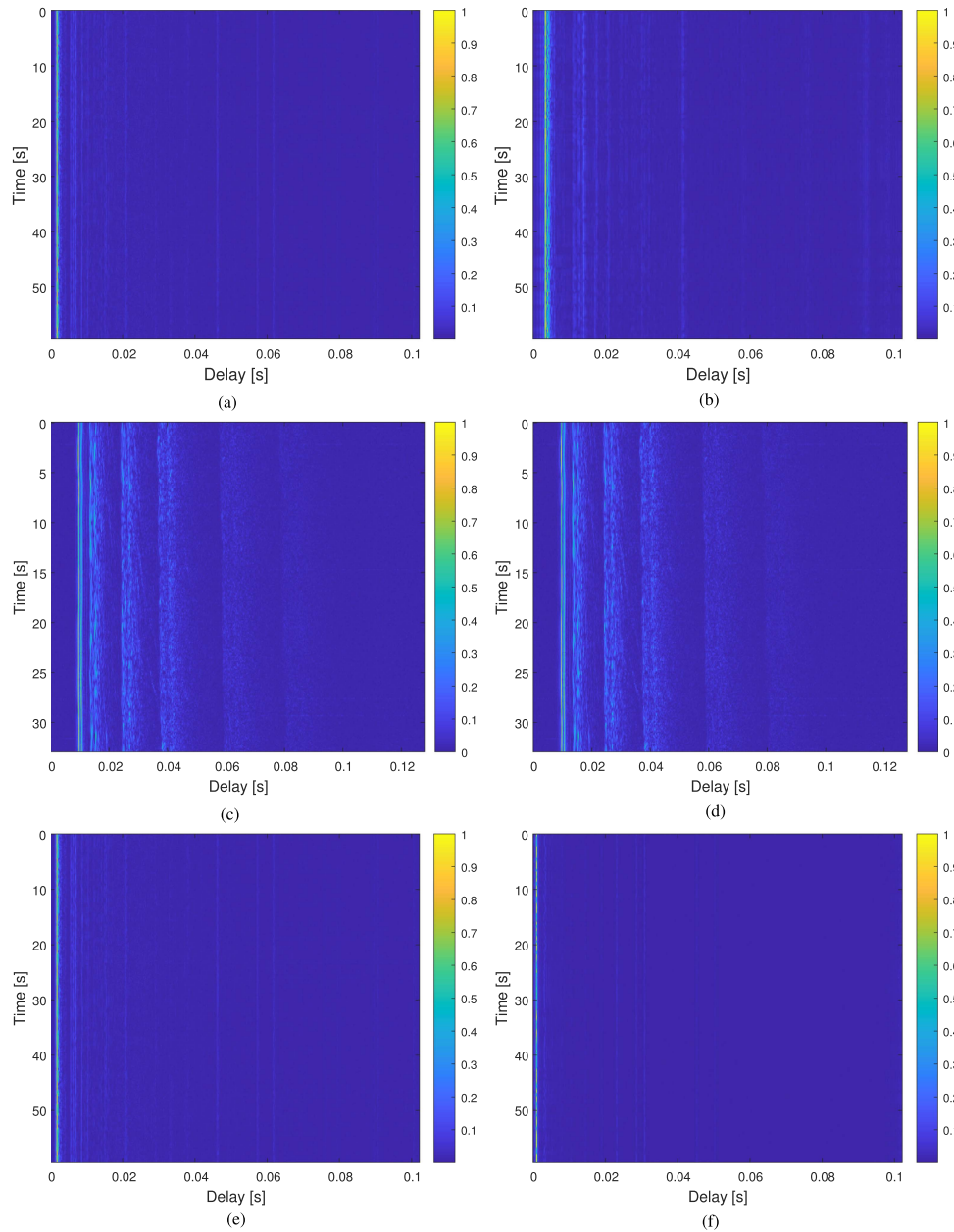


Fig. 5. DenseNet performance in the estimation of TVIR for BCH and KAU channels. (a) Actual TVIR of BCH channel. (b) Estimated TVIR of BCH channel. (c) Actual TVIR of KAU1 channel. (d) Estimated TVIR of KAU1 channel. (e) Actual TVIR of KAU2 channel. (f) Estimated TVIR of KAU2 channel.

algorithm shows marginal improvement in terms of BER across various SNR values. The FC-NN exhibits the curve that is most similar to our proposed model. Our proposed estimator, which is based on a deep neural network, exhibits superior performance compared to the LS algorithm and the FC-NN estimator across most SNR levels. Fig. 7 clearly demonstrates that the DenseNet estimator yields reduced BER outcomes compared to the other estimators across all channels.

While the DenseNet model demonstrates strong performance at higher SNR values, its performance at low SNRs, such as -5 dB, reveals some important nuances. At an SNR of -5 dB, DenseNet achieves BERs of approximately 0.0078, 0.013, and 0.0062 for the BCH, KAU1, and KAU2 channels, respectively,

when using BPSK modulation [Fig. 7(a)–(c)]. These values represent significant improvements over the LS estimator, which records BERs of approximately 0.015, 0.022, and 0.012 for the same channels, respectively, indicating reductions of 48%, 41%, and 48% in BER. The MMSE estimator shows BERs of 0.014, 0.021, and 0.0115, respectively. However, it is important to note that the FC-NN model slightly outperforms DenseNet at -5 dB, with BERs of 0.0075, 0.012, and 0.0060 for the BCH, KAU1, and KAU2 channels, respectively. In the case of QPSK modulation [Fig. 7(d)–(f)], DenseNet achieves BERs of approximately 0.011, 0.015, and 0.0078 at an SNR of -5 dB in the BCH, KAU1, and KAU2 channels, respectively. These values again demonstrate significant advantages over traditional methods,

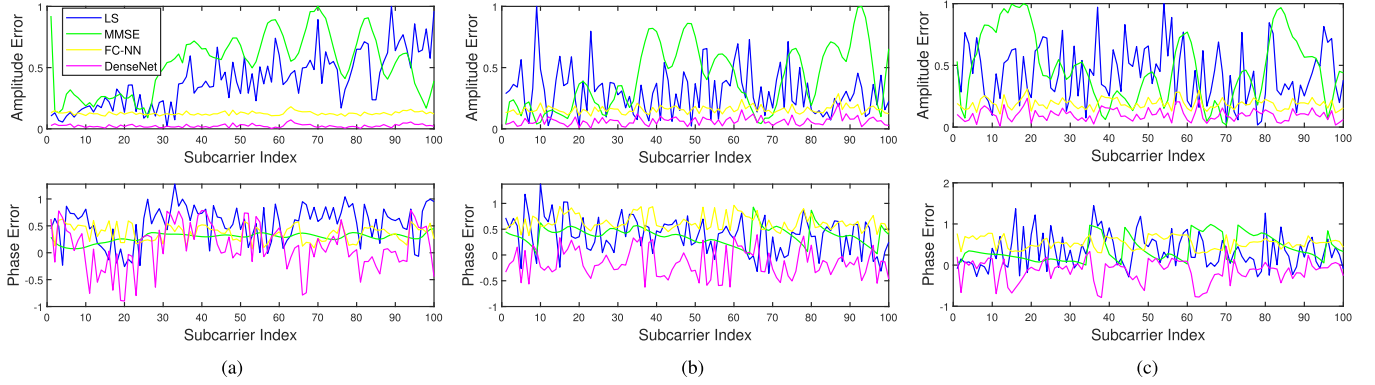


Fig. 6. Amplitude error and phase error in estimation of different channels. (a) Amplitude error and phase error of BCH. (b) Amplitude error and phase error of KAU1. (c) Amplitude error and phase error of KAU2.

where the LS estimator records BERs of approximately 0.021, 0.029, and 0.017, and the MMSE estimator records BERs of 0.019, 0.027, and 0.015, respectively. However, similar to the BPSK modulation case, the FC-NN model exhibits slightly better performance at -5 dB, with BERs of 0.0105, 0.0145, and 0.0075 for the respective channels.

Despite the slight edge that the FC-NN model has at -5 dB, DenseNet offers competitive BER reductions at low SNRs, which is crucial for practical applications where low SNR conditions are common. The ability of DenseNet to consistently outperform traditional methods like LS and MMSE, even at -5 dB, underscores its robustness and applicability in real-world scenarios. DenseNet shows competitive performance at higher SNRs and much better performance than all baseline methods including FC-NN. Although DenseNet shows a worse result than the worst FC-NN model at -5 dB, this must be seen in its general context. In practical scenarios, communication systems often operate across a range of SNR levels, not just at the extreme low end. DenseNet's superior performance at slightly higher SNR values, such as 0 dB and above, means that it can offer more reliable communication with fewer errors in many real-world situations. This broader SNR coverage guarantees that DenseNet remains more robust with consistent performance than baselines as conditions enhance, which is crucial in constantly dynamic and variable environments.

In addition, the architecture of DenseNet comes out to be cooperative for capturing complex relations within data which might result in better performance at high SNR levels. This capability can be leveraged in systems where maintaining low error rates is more critical than optimizing for the lowest possible SNR. While DenseNet already provides significant improvements over traditional methods, In future, further optimization and fine-tuning of its architecture could enhance its performance at very low SNRs like -5 dB. For example, adjustments to the training process, the introduction of more specialized layers, or the incorporation of domain-specific knowledge could help to close the gap between DenseNet and FC-NN at these challenging SNR levels. By focusing on these improvements, we believe that DenseNet can be further optimized with respect to the entire SNR spectrum. It can be seen that denseNet has better

TABLE V
AFR OF WATERMARK CHANNELS, CALCULATED BY USING EMD

Channels	P(d)	P(r)	P(h)	AFR
KAU	0.0011	0.0063	0.0074	0.8521
BCH	0.0016	0.0010	0.0026	0.3868

performance than conventional and imposes lower BERs for all SNR points, especially for 0 dB and above which is the region of interest for many practical systems. Hence, DenseNet's architecture can be further optimized to improve performance at the lowest SNRs, making it a promising candidate for future research and application in challenging environments.

4) *Robust Performance on Diverse Channels:* It is imperative to conduct an analysis of the various channels. This will not only enhance the understanding of UWA channels, but also facilitate a systematic evaluation of channel estimation performance. In [32], a detailed analysis was presented on the challenges associated with watermark channels. An illustration of how empirical mode decomposition (EMD) filtering is applied to BCH and KAU channels can be seen in Table V. EMD filtering is pivotal in this research as it has allowed scholars to design a way to evaluate individual UWA channels by determining the average fade rate, a metric that is exhibited in Table IV.

EMD possesses the unique capability to separate a signal into trend signal and random signal. When considering the UWA channel, it is conceived as a blend of both slow and fast fading. This is evident when each tap of the channel is denoted as $h_i(n) = d_i(n) + r_i(n)$. Here, $d_i(n)$ epitomizes the deterministic or what can be referred to as the "trend" segment of the channel, whereas $r_i(n)$ denotes its random component. Intriguingly, this deterministic portion is labeled "pseudodeterministic" [32].

Further diving into the mechanics, each channel tap, represented as $h_i(n)$, can undergo decomposition via EMD. This results in a sum of modes $m_{i,q}(n)$ accompanied by a residue $e_i(n)$. A pivotal metric called the average fade rate (AFR) emerges from this, offering a glimpse into the quality of the channel. This rate is ascertained from the potency of the random

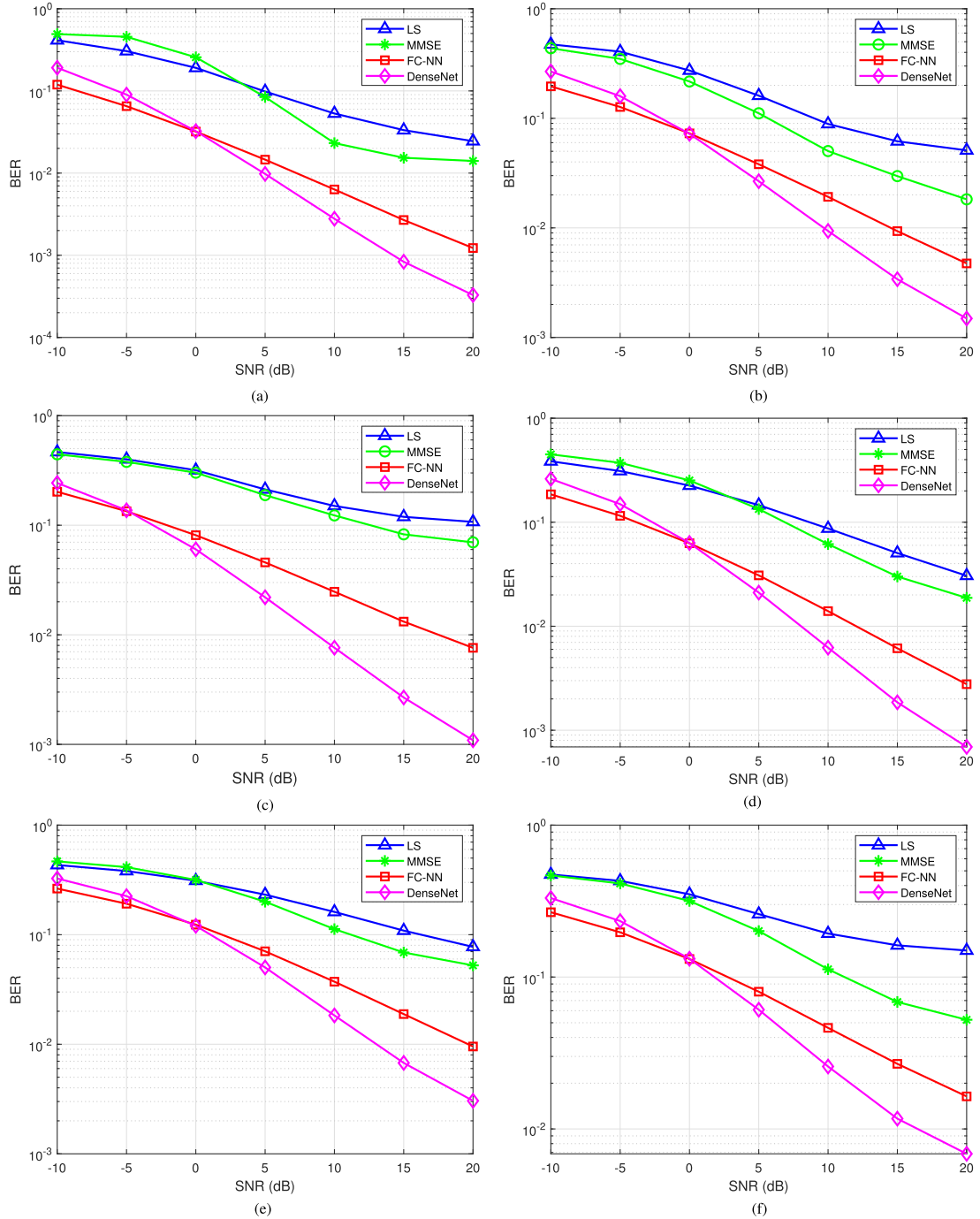


Fig. 7. Impact of modulation scheme. (a) The BPSK modulation using BCH channel. (b) The BPSK modulation using KAU1 channel. (c) The BPSK modulation using KAU2 channel. (d) The QPSK modulation using BCH channel. (e) The QPSK modulation using KAU1 channel. (f) The QPSK modulation using KAU2 channel.

segment $r_i(n)$ in relation to the channel $h_i(n)$. The AFR in [32] is defined as

$$AFR_R = 10 \log \left(\frac{P_{ow}(r)}{P_{ow}(h)} \right) = 10 \log \left(\frac{\sum_{n=1}^N \sum_{i=1}^I \|r_i(n)\|^2}{\sum_{n=1}^N \sum_{i=1}^I \|h_i(n)\|^2} \right) \quad (34)$$

where

- 1) $Pow(r)$ is the power of the random component $r_i(n)$.
- 2) $Pow(h)$ is the power of the total channel $h_i(n)$.
- 3) $Pow(d)$ is the power deterministic or trend component $d_i(n)$.
- 4) N is the total number of samples.
- 5) I is the number of channel taps.

This analysis suggest that BCH channel exhibit favorable quality as they possess stable paths that efficiently transmit a significant portion of the received signal energy. Nevertheless,

channels with high AFR values, such as KAU, pose greater difficulties due to their abundance of distinct trailing paths and unpredictable arrival patterns.

Furthermore, as depicted in the Fig. 7, the disparities in BER performance among different environments, namely KAU and BCH, can be attributed to the spatial and temporal characteristics of UWA channels. Spatially, channels like BCH, which demonstrate lower BER, are typically characterized by more stable propagation paths that maintain a consistent signal quality over distance. Temporally, these channels are less susceptible to rapid changes over time, which contributes to a more reliable signal. Conversely, environments like KAU, with higher BER, are affected by spatially diverse and temporally fluctuating paths. The quantitative evaluation of UWA channels in the Table V is strongly supported by these findings. To provide precise details, it can be observed from Fig. 7 that there exists a positive numerical correlation between the BER performance of the model and the AFR in every UWA environment. The exceptional performance of the proposed DenseNet estimator is observed exclusively when the pilot information is available.

DenseNet does not explicitly assume UWA channel statistics and noise variance, in contrast to the MMSE estimator, which is limited by prior knowledge of SNR. However, the data-driven training approach incorporates these factors implicitly into the DenseNet. Therefore, it can be regarded as the optimal selection in real-world scenarios.

Table V provides a performance comparison in terms of BER among four methods—LS, MMSE, FC-NN, and DenseNet for BPSK and QPSK modulation schemes. For BPSK with the BCH category, DenseNet yields the lowest BER at 0.0006, followed closely by FC-NN at 0.0010, while LS has a BER of 0.04001 and MMSE registers 0.03020. In the context of QPSK under the KAU2 category, DenseNet similarly excels with a BER of 0.0080, whereas LS records the highest error rate at 0.3001. MMSE and FC-NN show intermediate performance levels. Cumulatively, DenseNet consistently emerges as the top-performing method in minimizing BER across both modulation types and various categories.

Table V shows that performance of the model experiences a substantial decline when we apply it to KAU1 and KAU2. The performance gap observed can be attributed to the inability of the model to effectively adapt the knowledge acquired in a channel with a relatively simple structure (BCH) to a more complex environment (KAU1 and KAU2). Furthermore, few-shot learning has the potential to serve as a valuable tool for addressing scenarios with limited data availability or mismatches in the communication environment in real-world applications.

5) *Evaluation on Impact of Pilot Number*: The number of pilots plays an important role as a supplement to the theoretical analysis. In the proposed model, the correlation between the BER performance and the number of pilots is crucial. The introduction of pilots is essential for accurate channel estimation, necessitating a strategy that uses the minimum number of pilots required. Such an approach enhances spectrum utilization and efficiency without compromising the system's overall performance.

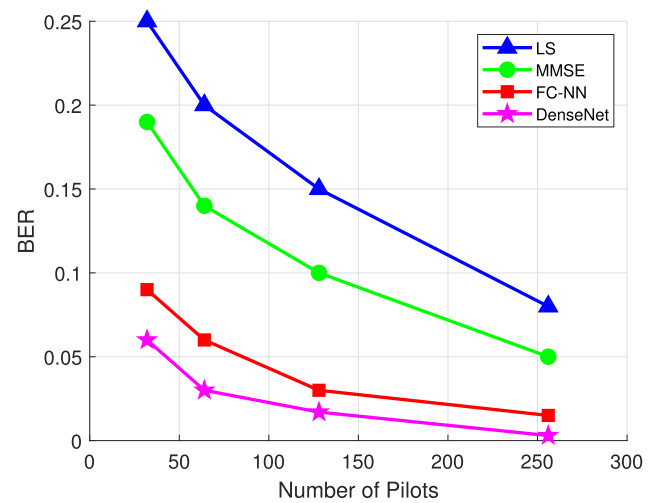


Fig. 8. Impact on pilot number.

As shown in Fig. 8, when the number of pilots is increased from 16, 32, and 256, the BER performance of all pilot-assisted methods, with the exception of FC-NN and DenseNet, reduce significantly. Fortunately, despite the reduction in pilot tone, the DenseNet estimator can still sustain its BER to nearly 10^{-3} in BCH, whereas the LS and MMSE estimators fail with a BER of approximately 10^{-2} .

Furthermore, it is evident that the benchmark estimators fail to converge to the DenseNet estimator as the number of pilots decreases. According to the data presented in Fig. 7, the DenseNet estimator demonstrates superior performance compared to other estimators across all channels, namely BCH, KAU1, and KAU2, for both modulation schemes employed. The design of the DenseNet estimator is inherently geared towards harnessing the complete spectrum of channel information, which explains its marked performance dip in scenarios with insufficient pilot signals. In contrast, the MMSE estimator maintains a certain level of performance enhancement due to its ability to capitalize on the pilot signals present. Based on the aforementioned analyses, it can be observed that the estimator gradually converges towards the DenseNet estimator in an asymptotic manner as the number of pilots increase. Therefore, the presence of an adequate number of pilots within the transmitted OFDM symbols is a necessary and sufficient condition. Overall, the BER performance over varying SNR remains within an acceptable threshold, which demonstrates a significant advantage of the proposed method as appropriate pilot estimation method. This approach exhibits a certain degree of adaptability in handling fluctuations in the necessary quantity of pilots for signal retrieval.

E. Robustness Analysis Under UWA Environment Variability

As highlighted in the article, UWA channels exhibit significant temporal and spatial variability, posing a challenge for the robustness and adaptability of DL models. To rigorously evaluate the DenseNet model's ability to generalize to unseen data, an essential capability for real world applications, we

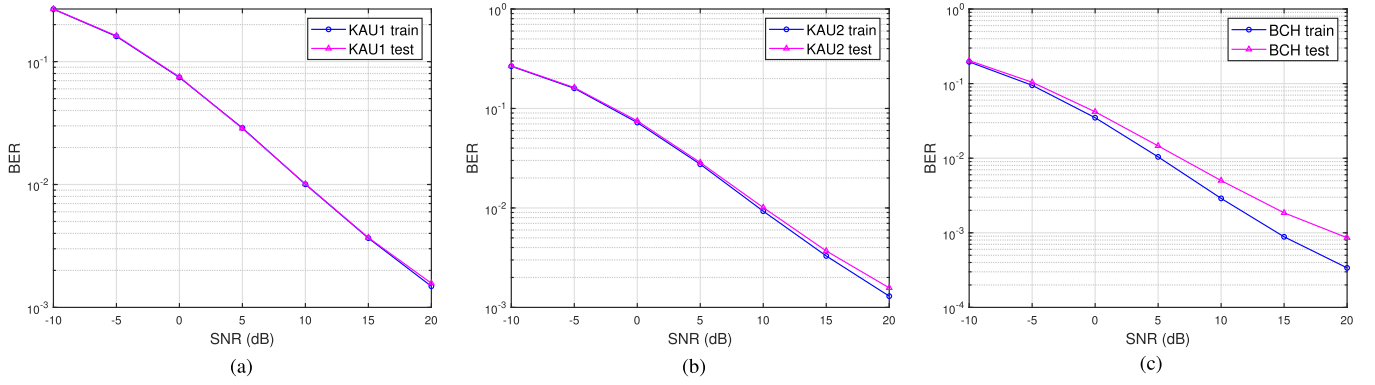


Fig. 9. BER performance on training and unseen data. (a) The BER performance for KAU1 channel on training and unseen test data channel. (b) The BER performance for KAU2 channel on training and unseen test data channel. (c) The BER performance for BCH channel on training and unseen test data channel.

TABLE VI
DENSENET COMPLEXITY

Component	Complexity
DenseBlock (1)	$O(K_c \times C_{in}) + O(L_f \times K_c^2) + O(P_c)$
DenseBlock (2)	$O(K_c \times C_{in}) + O(L_f \times K_c^2) + O(P_c)$
Transition Layer (1)	$O(P_c)$
Transition Layer (2)	$O(P_c)$
Fully Connected Layer	$O(L_f \times K_c^2)$
Input Data Overhead	$O(N \times M)$

TABLE VII
COMPLEXITY COMPARISON

Algorithm	Complexity
LS [3]	$O(K_c^2)$
MMSE [3]	$O(K_c^3)$
CNN [17]	$3(O(L_c \times K_c \times K_s \times C_{in} \times C_{out}) + 2(L_f \times K_c^2))$
FC-NN [11]	$5(L_f \times K_c^2)$
Proposed DenseNet	$2 \times (O(K_c \times C_{in}) + 2 \times O(P_c)) + O(L_f \times K_c^2)$

conducted additional experiments to test the model's performance under conditions it had not encountered during training. The data set comprises three distinct channel soundings: BCH, KAU1, and KAU2, with BCH consisting of four different TVIRs and KAU1 and KAU2 each consisting of 16 different TVIRs. These CIRs were derived from TVIRs measurements taken at different times and under varying environmental conditions, thereby capturing the inherent temporal and spatial variability of the underwater acoustic channels. As mentioned earlier in the DNN-based computation framework, we took 50% of the CIRs from each channel as training set for network training and kept the other 50%, which are completely unseen during the training phase as test set to compute performance. This approach ensures that the model is evaluated on "unfamiliar" test samples, effectively simulating real-world scenarios where the network encounters novel environmental conditions. This strategy offers an authentic insight into the model's ability to generalize and withstand variations in real-world situations.

The BER performance of the DenseNet model on the KAU1 and BCH channels is depicted in Fig. 9. For the KAU1 and KAU2 channel, as shown in Fig. 9(a) and (b), the BER curves for both the training and unseen test data align closely across various SNRs. At an SNR of 0 dB, the BER for the training data is approximately $10^{-1.5}$, while the BER for the unseen test data is $10^{-1.4}$, indicating a minor difference of about 7%. This close alignment continues across higher SNR levels, with the BER performance on the unseen test data being within 5% of the training data across all SNR levels. This demonstrates that the model maintains its performance even when exposed

TABLE VIII
PERFORMANCE ANALYSIS IN TERMS OF BER

Method	BPSK			QPSK		
	BCH	KAU1	KAU2	BCH	KAU1	KAU2
LS	0.0401	0.0800	0.0101	0.0100	0.0101	0.3020
MMSE	0.0301	0.0800	0.1001	0.0401	0.0902	0.2010
FC-NN	0.0010	0.0070	0.0091	0.0052	0.0101	0.0401
DenseNet	0.0006	0.0040	0.0011	0.0011	0.0050	0.0080

to unseen samples from the KAU1 environment, showcasing high robustness and adaptability. For the BCH channel, Fig. 9(c) reveals a slight deviation between the BER curves for the training and test data, particularly at higher SNRs. At an SNR of 10 dB, the BER for the training data is 10^{-3} , while the BER for the unseen test data is approximately $10^{-2.8}$, representing a marginal performance gap of about 6.3%. Despite this marginal deviation, the performance gap remains relatively small, and the model still achieves good results on the unseen test samples. The BER performance on the unseen test data was within 10% of the training data at higher SNR levels, indicating a manageable decline in performance. This slight increase in BER for the test data at higher SNRs can be attributed to the complex nature of the BCH channel, which presents a more challenging environment for the model to adapt to.

Overall, these additional experiments confirm that the DenseNet model exhibits strong generalization capabilities

TABLE IX
MAPPING OF PROBLEMS ADDRESSED, PROPOSED SOLUTIONS, AND VALIDATION

Problem addressed	Proposed Solution and Research Contributions	Validation
Challenge: Need for Prior Statistics and Lack of generalization for complex channel conditions	- The proposed DenseNet-based channel estimator, demonstrating generalization in handling diverse underwater channel conditions. - DenseNet estimator provides a mathematical model to explain its behavior. - Eliminates the need for complex channel characteristics like SNR	Validation through extensive simulations and real-world underwater data (Fig. 6), showing the generalization and effectiveness of the DenseNet estimator.
Challenge: Lack of Robustness on Different Underwater Channels	- Confirming the robustness of the DenseNet-based estimator across diverse underwater channels, including BCH and KAU channels, by maintaining reliable performance and adapting to diverse scenarios. - The model is trained using real-time varying CIR from the WATERMARK dataset, ensuring diversity for accurate channel estimation.	Validation of robustness across different underwater channels through experimental results and comparisons (Fig. 7), demonstrating the model's adaptability to varying channel conditions.
Issue: Enhancing efficiency by minimizing pilot overhead in underwater channel estimation	- Assessment of the impact of pilot symbols on the performance of the DenseNet estimator, demonstrating the model's ability to achieve acceptable BER levels with a reduced number of pilot resources. - Reduction in pilot signals minimizes bandwidth overhead in practical applications.	Validation of pilot overhead reduction through numerical results and BER comparisons with other models (Fig. 8).

and maintains robust performance when tested on previously unseen data, representing 50% of the channel samples. Training on a subset of the data and testing on the remaining samples provides a more realistic evaluation of the model's adaptability in practical applications. These findings underscore the effectiveness of the DenseNet model in handling the inherent variability of underwater acoustic channels, ensuring reliable performance in real-world scenarios. Future research will explore few-shot learning techniques to further enhance the model's adaptability in data-scarce situations and environments with significant variability between training and testing stages.

F. Model Complexity Analysis

The computational complexity of a proposed DenseNet model is assessed in comparison to alternative algorithms such as LS, MMSE, and FC-NN estimators. This complexity analysis elucidates the computational demands of these methods. As indicated in Table VII, the LS algorithm presents the simplest complexity of $O(K_c^2)$, where K_c signifies the number of subcarriers [32]. However, the LS algorithm's performance is relatively constrained in contrast to other methodologies. Conversely, the MMSE approach, though theoretically optimal, necessitates substantial computational resources due to its $O(K_c^3)$ complexity, predominantly due to matrix inversion and scalar multiplication. Furthermore, the complexity of the CNN estimator incorporates both Convolutional (Conv) layers $O(K_c \times K_s \times C_{in} \times C_{out})$ and FC layers $O(K_c^2)$. For a CNN with L_c convolutional layers and L_f FC layers, the overall complexity becomes $O(L_c \times K_c \times K_s \times C_{in} \times C_{out} + L_f \times K_c^2)$. The computational nature may still vary with increasing

convolutional layers and FC layers. Specifically, the computational complexity per FC layer is $O(K_c^2)$, where K_c denotes the subcarrier count. For an L -layer DNN with L considerably smaller than K_c , the overall complexity is $O(L \times K_c^2)$.

In terms of computational complexity, the LS and MMSE algorithms are considerably less demanding than neural network-based approaches like CNN [17], proposed DenseNet, and FC-NN. Specifically, LS has a complexity of $O(K_c^2)$ and MMSE has $O(K_c^3)$, making them both polynomially dependent on the number of subcarriers K_c . Neural network-based methods introduce additional variables such as the number of convolutional layers L_c , kernel size K_s , and the number of input and output channels, making them significantly more computationally complex.

where

- 1) K_c is the number of subcarriers.
- 2) K_s is the kernel size in Convolutional layers.
- 3) C_{in} is the number of input channels to Convolutional layers.
- 4) C_{out} is the number of output channels from Convolutional layers.
- 5) L_c is the number of Convolutional layers in the CNN model.
- 6) L_f is the number of FC layers in the CNN model.
- 7) L is the number of layers in the DNN model.
- 8) N and M are the input data dimensions for DenseNet model.

The complexity analysis of the DenseNet is shown in Table VI. The table outlines the computational complexity of a DenseNet architecture, breaking it down into its key components. Each DenseBlock involves a convolutional layer, FC layer, and average pooling with respective complexities. Transition layers add their complexity due to pooling. The FC layer contributes

to the overall complexity, and there is additional overhead for input data dimensions. This comprehensive breakdown assists in assessing the computational demands of the DenseNet model, aiding in resource allocation. The DenseNet architecture strategically balances computational complexity by integrating a pair of DenseBlocks, each with convolution layers and pooling operations. This design ensures its applicability to DL tasks without excessive computational burden. The selection of key parameters namely, the number of subcarriers (K_c), the number of input channels (C_{in}), the number of layers in the FC layer (L_f), and the pooling size (P_c) follows a heuristic based on the specific requirements of the task.

Table VIII presents performance analysis in terms of BER for various modulation methods (BPSK and QPSK) on different underwater channels (BCH, KAU1, and KAU2). It shows the BER values for each method and modulation-channel combination, offering a comparative view of their performance. Table IX serves as a mapping of problems addressed, proposed solutions, and validation in the context of underwater channel estimation. It outlines the challenges related to the need for prior statistics and generalization in complex channel conditions and the lack of robustness on different underwater channels. The proposed solutions, such as the introduction of the DenseNet-based channel estimator and assessment of pilot overhead, are detailed. The validation of these solutions through experimental results and real-world data are also summarized.

V. CONCLUSION

This study comprehensively tackled the complex channel estimation problem in UWA OFDM communication system. We have studied how different modulation schemes, different WATERMARK channels, and the number of pilots affect channel estimation approaches. Our unique DenseNet-based estimator is designed to handle UWA channel complexity. We explained the DenseNet model's architecture and training. Through extensive experiments utilizing the WATERMARK data set, drawn from real-world underwater environment, we demonstrate how the DenseNet estimator works. Experiments show that DenseNet is better than classic approaches. The proposed DenseNet model outperforms LS and MMSE and FC-NN estimators, improving BER by 96.3%, 94.2%, and 40.0%, respectively. Our experiments show that BPSK performs better than QPSK in modulation schemes. This shows the necessity of choosing the right modulation technique for UWA OFDM systems. The performance examination of our model over multiple underwater channels shows its robustness in both stable as well as demanding conditions. The DenseNet-based estimator outperforms conventional approaches due to its versatility. Interestingly, our study examines how pilot signals affect the DenseNet estimator's performance. Our model maintains acceptable BER levels with fewer pilot resources, which is important for underwater communication applications. DenseNet-based estimator has shown its adaptability, robustness, and higher performance over existing channel estimate methods, making it a major advance in underwater communication systems. The DenseNet model may improve underwater communication networks' reliability

and efficiency by tackling the particular problems of this dynamic and unpredictable environment. In future, we plan to do experiments on real-world sea data for comprehensive testing, to enhance the model's generalization capabilities across a wider range of UWA environments.

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